Perception-Driven Robotic Gameplay using YOLO Detection and Dobot Magician Lite

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ABSTRACT

This lab demonstrates the integration of perception, decision making and actuation through a human-robot interactive Tic-Tac-Toe game using the Dobot magician lite robotic arm. This system uses a YOLO- based symbol detection model via the Roboflow API to identify the human player moves. A python-based Minimax Algorithm determines the robot optimal response, which the Dobot executes by actually drawing the grid. This lab provides hand-on approach on integrating perception using python, open-cv and Dobot SDK.

1. INTRODUCTION

The purpose of this lab was to design a real-time interactive robotic system that combines computer vision and robotic control for gameplay automation. Using the Dobot magician lite, a 3*3 Tic-Tac-toe grid was drawn and controlled , while a webcam and YOLO-model detected the players moves. The robot was then autonomously computed and executed its next move based on the minimax algorithm. This explores the fundamentals of AI-based decision-making, coordinate mapping and robotic actuation.

2. PROBLEM STATEMENT

This lab requires to develop a system that:

1. Detect player symbol(X/O) through camera input and yolo based object detection.
2. Records the player's moves in real time and interprets board state.
3. Uses a minimax algorithm to compute the robots optimal move.
4. Commands the Dobot Magician Lite to physically draw the corresponding symbol on the board.
5. Displays the game result via robot-drawn indicators.
6. Checks for error detection, if human tries to cheat by drawing twice or drawing the wrong symbol.

3. METHODOLOGY

A. SYSTEM ARCHITECTURE

The system consists of two synchronized scripts:

- Detection.py: Runs YOLO- based detection using Roboflow API to identify "X" and "O" positions.
- Midsem.py: manages game logic, robot control, and AI using the pydobot library.
- Abc.txt: Data exchange occurs via a text file where decision results are logged from detection.py and read by midsem.py.

B. VISION SYSTEM

- **Model:** Custom YOLOv8 model trained on Tic-Tac-Toe symbols.
- **Input:** Real-time video stream from a webcam.
- **Processing:** Bounding boxes are analysed, and each detected symbol is mapped to its respective grid cell.
- **Output:** Writes coordinates and symbols in text format for robot processing.

C. ROBOT CONTROL

- Robot: Magician lite with pen attachment/

- Motion Commands: Executed via pydobot library using cartesian coordinates.
- Z_safe: -30 mm(travel height)
- Symbols:
  - X drawn with intersecting diagonals.
  - O drawn via circular interpolation.

D. GAME LOGIC

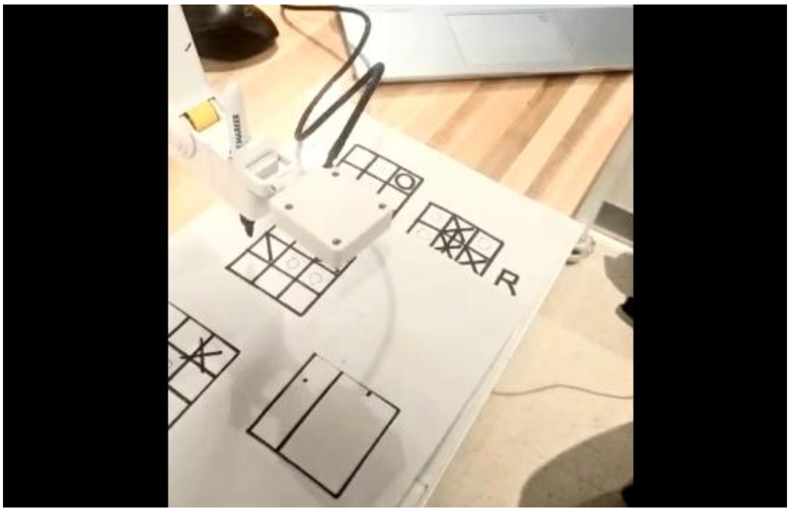
- The Minimax Algorithm is implemented to determine optimal AI moves.
- The robot alternated turns with the human, checking for winners or Tie after every move.
- Winning conditions are automatically verified through row, column or diagnol checks

E. WORKFLOW

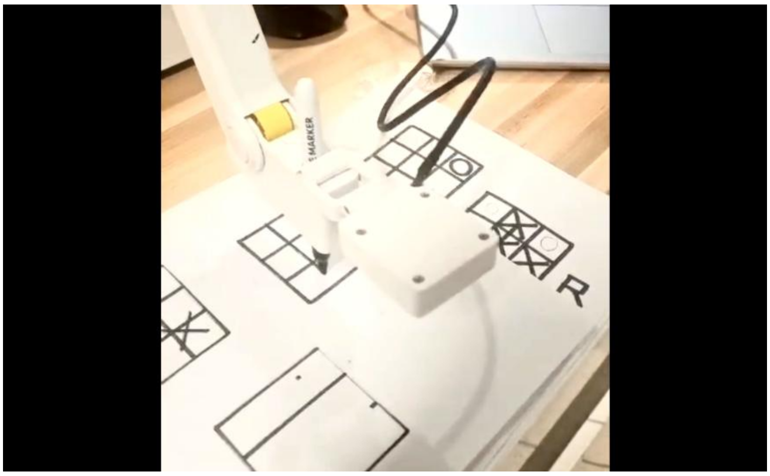
- Robot draws the initial grid.
- Camera detects player's symbol and writes it to abc.txt.
- Main program updates board state.
- AI computes next move → Robot executes drawing.
- The process repeats until a win or tie is detected.

4. RESULTS

A. Making the grid



B. Robot playing X and O



C. Performance Metrics

- Detection Accuracy: ~93%
- Average cycle time per move: 10 seconds
- Response delay: 3 seconds
- Total playtime: 2.5 minutes per game

5. REFELCTIONS

- Learned integration of **AI logic and robotic actuation** in a single pipeline.
- Understood real-time synchronization between **vision inference** and **hardware execution**.
- Calibrating robot coordinates with camera field-of-view was essential for accuracy.
- Developed debugging skills for Python multiprocessing and serial communication errors.
- Gained insights into human–robot collaboration through gamified interaction.

6. CONCLUSIONS

This experiment successfully implemented a complete closed-loop interactive system combining perception, AI decision-making, and robot control. The Dobot Magician Lite effectively interacted with the player using camera-based symbol detection and autonomous move computation.
This setup not only demonstrates the potential of low-cost educational robots for AI-embedded tasks but also provides a foundation for industrial applications such as **vision-based sorting**, **gesture-based controls**, and **autonomous decision-making systems**.

Serial Number of the Robot:
DT-MGL-MR002-01E

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APPENDIX:

- Detection.py
- Midsem.py
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